TARNDEEP SINGH

tarndeep452@gmail.com | +91 8699242254 | LinkedIn | GitHub

SUMMARY

4th-year B.Tech Computer Science and Engineering student specializing in Data Science with a CGPA of 9.02 and hands-on experience in Python, SQL, machine learning, data preprocessing, feature extraction, and model evaluation. Developing an end-to-end Indian Sign Language recognition system involving dataset analysis, landmark-based feature extraction, temporal preprocessing, data augmentation, and deep learning. Seeking a Data Science internship focused on machine learning, data analysis, and building data-driven solutions.

EDUCATION

Akal University — B.Tech CSE (Data Science) Aug. 2023 – Jun. 2027
CGPA: 9.02/10.0 Talwandi Sabo, Punjab

XII (CBSE) | Akal Academy Rampur Narotampur 2023 | 81%
X (CBSE) | Akal Academy Rampur Narotampur 2021 | 95%

PROJECTS

Indian Sign Language (ISL) Recognition | GitHub Ongoing

- Developing an end-to-end machine learning pipeline for Indian Sign Language recognition, including dataset analysis, preprocessing, feature extraction, sequence preparation, model training, and evaluation.
- Using MediaPipe Holistic to extract pose, hand, and face landmarks and transform video sequences into structured numerical features for deep learning models.
- Evaluating temporal preprocessing, data augmentation, and class balancing strategies to improve model generalization across gesture classes.
- **Technologies:** Python, TensorFlow, MediaPipe, OpenCV, NumPy, Pandas, Machine Learning

FocusLab: Study Session Tracker and Analytics | GitHub September 2026

- Built a web-based study tracker, extended from a Python Pomodoro timer, that logs focus sessions by subject, duration, and completion status with persistent storage in SQLite.
- Developed an analytics dashboard showing weekly study trends, subject-wise time distribution, and streaks using Pandas and Plotly; deployed as a public web app for student use.
- **Technologies:** Python, Streamlit, SQLite, Pandas, Plotly

TECHNICAL SKILLS

Programming Languages: Python, Java, SQL
Machine Learning: Computer Vision, Feature Extraction, Data Preprocessing, Model Evaluation, Scikit-learn, TensorFlow
Data Science: NumPy, Pandas, Matplotlib, Seaborn, Data Analysis, Data Visualization
Databases and Tools: SQL, Git, GitHub, VS Code, PyCharm, Jupyter Notebook, Google Colab
Core Computer Science: Data Structures and Algorithms, Object-Oriented Programming, Problem Solving, Database Management Systems

ACHIEVEMENTS

- Led the planning and execution of **TechTrail '26**, a campus-wide Treasure Hunt for **99 participants across 33 teams**, designing riddles, location clues, decoding challenges, checkpoints, and the final treasure sequence.
- Selected for the Internal Evaluation Round of **Smart India Hackathon (SIH) 2025**.
- Secured **1st position** in a Cybersecurity Quiz among 10 participating teams in 2025.
- Qualified through two rounds of an **individual coding hackathon** conducted by CodeForge Club in 2026.

LEADERSHIP AND VOLUNTEERING

DataNinjas – Founding President | Akal Tech Society, Akal University Nov. 2024 – Dec. 2025

- Helped establish and build a data science focused technical club to a 75 member community, driving technical engagement and peer learning.
- Planned and executed technical competitions including Quiz (48 participants), Debate (25 participants), and Treasure Hunt (90 participants across 30 teams).
- Designed logic and cipher based Treasure Hunt challenges and coordinated volunteers, teams, and event logistics.

FutureLens – Vice President | Akal Tech Society, Akal University Apr. 2026 – May 2026

- Coordinated debate and presentation competitions and collaborated with club leadership and technical and communication teams under Turing Truncheons — Akal Tech Society.

CI2A 2026 – Conference Volunteer | Akal University Apr. 2026

- Supported backend coordination, session management, documentation, and communication with faculty, researchers, and the organizing team.

RELEVANT COURSEWORK

Data Structures and Algorithms, Object-Oriented Programming, Database Management Systems, Data Handling and Visualization, Statistics, Machine Learning, Artificial Intelligence, Deep Learning, Natural Language Processing